RAJAGIRI SCHOOL OF ENGINEERING & TECHNOLOGY (AUTONOMOUS)

Department of Artificial Intelligence & Data Science

102008/AD701S – BIG DATA ANALYTICS LAB – LAB EXAMINATION

QUESTION PAPER – SET A

Topics: Hadoop HDFS • MapReduce • Hive • Spark

Instructions

- Answer any three questions. Each question has two parts, (a) and (b).
- The data files (orders_jan.csv, orders_feb.csv, access_log.csv, employees.csv, online_orders.csv) are provided separately.

Question 1 – Hadoop HDFS

Data file(s): `orders_jan.csv, orders_feb.csv`

(a) Using hdfs dfs commands only, create /user/student/setA/incoming and /user/student/setA/processed in a single command, upload both order files into incoming, copy orders_jan.csv into processed under the name jan_done.csv, display the total space used by /user/student/setA, and finally delete incoming and show a recursive listing of /user/student/setA.

(b) Write a Python program using the hdfs library that uploads both order files into /user/student/setA_py, reads each file directly from HDFS to find its number of orders and total amount, writes these figures as summary.csv into the same HDFS directory, and downloads summary.csv to the local home folder.

Question 2 – MapReduce

Data file(s): `access_log.csv`

(a) Using Hadoop Streaming in Python, count the number of requests made by each IP address and output only the IP addresses with at least 3 requests, skipping the header and malformed lines. Test locally with a Unix pipe and then run the job on Hadoop.

(b) Using Hadoop Streaming in Python, output for every URL the number of hits, the total bytes served and the number of error responses (status 400 or above). Test locally with a Unix pipe and then run the job on Hadoop.

Question 3 – Hive

Data file(s): `employees.csv`

(a) Create a database company_exam with a table employees matching employees.csv (skipping the header row), load the file, and write a query that shows, for each department, the number of employees, the average salary rounded to 2 decimals and the maximum salary, ordered by average salary from highest to lowest.

(b) Write a HiveQL query that lists the cities having at least two employees who joined before the year 2020, along with that number of employees.

Question 4 – Spark

Data file(s): `online_orders.csv`

(a) Write a PySpark program that loads online_orders.csv, converts the numeric columns to proper types, removes duplicate rows and orders without a customer, fills a missing quantity with 1 and a missing unit_price with the average unit price (2 decimals), adds a total_amount column (quantity × unit_price), and saves the cleaned data to ~/orders_clean.

(b) Register the cleaned data as a temporary view named orders and use Spark SQL to find the number of orders and total revenue per category, highest revenue first; then produce the same result using the DataFrame API.

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102008/AD701S – BIG DATA ANALYTICS LAB – LAB EXAMINATION

QUESTION PAPER – SET B

Topics: Hadoop HDFS • MapReduce • MongoDB • Spark

Instructions

- Answer any three questions. Each question has two parts, (a) and (b).
- The data files (morning.txt, evening.txt, extra.txt, math.csv, physics.csv, chemistry.csv, weather.csv, books.csv, patients.csv) are provided separately.

Question 1 – Hadoop HDFS

Data file(s): `morning.txt, evening.txt, extra.txt, math.csv, physics.csv, chemistry.csv`

(a) Using hdfs dfs commands only, create /user/student/setB/logs, upload morning.txt and evening.txt into it, append the contents of extra.txt to the end of morning.txt in HDFS, merge all files in the directory into a single local file ~/all_logs.txt and display it, and finally delete /user/student/setB and confirm the deletion.

(b) Write a Python program using the hdfs library that uploads the three mark files from ~/marks_local into /user/student/setB_marks/raw, reads each file from HDFS to find the subject average and topper, writes this report to /user/student/setB_marks/report/subject_report.txt, and then moves the files from raw to /user/student/setB_marks/processed.

Question 2 – MapReduce

Data file(s): `weather.csv`

(a) Using Hadoop Streaming in Python, output for each city the maximum, minimum and average temperature (2 decimals) and the number of valid readings, skipping the header and invalid (NA) readings. Test locally with a Unix pipe and then run the job on Hadoop.

(b) Using Hadoop Streaming in Python, output for each date the city that recorded the highest temperature on that date, along with the temperature. Test locally with a Unix pipe and then run the job on Hadoop.

Question 3 – MongoDB

Data file(s): `books.csv`

(a) Write a PyMongo program that loads books.csv into the collection books of the database libraryexam (numeric fields stored as numbers), counts the Fiction books, lists the 3 most recently published books, and displays page 2 of the catalogue when it is sorted by price with 4 books per page.

(b) Write a PyMongo aggregation pipeline that considers only books published in 2018 or later and computes, for each genre, the number of books, the average price and the total number of copies, sorted by total copies in descending order.

Question 4 – Spark

Data file(s): `patients.csv`

(a) Write a PySpark program that loads patients.csv, converts the numeric columns to proper types, removes duplicates, fills missing age and bill_amount with their averages (age as a whole number, bill to 2 decimals) and a missing ward with "General", converts gender to upper case, adds an age_group column (Child below 18, Adult 18–59, Senior 60+), and saves the result to ~/patients_clean.

(b) Register the cleaned data as a temporary view named patients and use Spark SQL to find the number of patients, average bill and average days admitted for each age group; then produce the same result using the DataFrame API.